COLLAPSE THEORY · SIMULATION 3

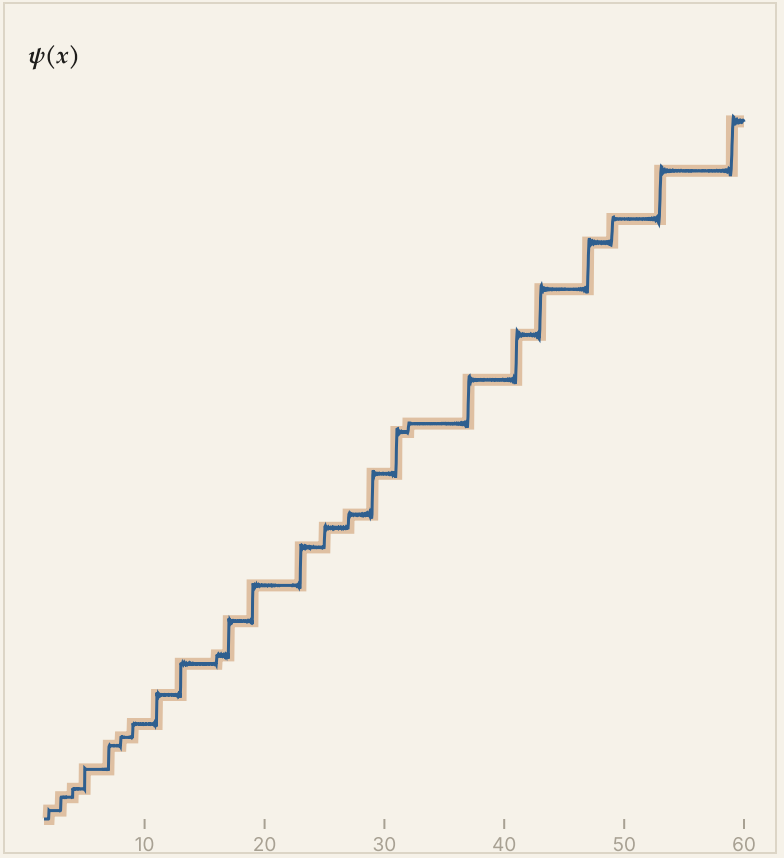
The Prime Oasis

Can a system of potential predict primes?

Day 325.

The engine is given multiplication and nothing else. It finds the primes as its own collapses, breaks on $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, and runs out of congruence laws on day 325. It climbs to a polynomial, then to a product of Euler's pentagonal series that names every prime $x^2 + 27y^2$ below a million without one error. Then it finds one such rule for every way to split 24.

17,981 collapses · 70,266 algebraic integers · 941,939 prime checks · 10 blind forecasts to 10^{16}



The first 1,000 zeros of zeta, added as waves,
draw the prime staircase. Orange band: the staircase itself.

Art Shirk

THE SETUP

The third Oasis runs on multiplication

	Number Oasis	Logic's Oasis	Prime Oasis
Desert	every length the tools can draw	every Turing machine with n states	**whole numbers, then Z[√−5], then primes**
Inhabitants	measurers, each holding a length	machines, small instruction tables	**numbers, arriving in order of size**
Command	"Measure!"	"Halt, or run forever?"	**"Be built from generators, one way."**
Constitution	every length is a ratio	rules that never answer wrongly	**generators; later, a law about each prime**
Growth	one measurer per day	one more state per day	**one newcomer per day from day 21**
Absurd potential	pairs that never finish measuring	machines no rule can settle	**newcomers it cannot build or predict**
Logical collapse	the cut, a new kind of number	a new rule that settles them	**a new generator, ideal, or law**
Collapse check	test against every known fact	the same test	**C1, C2, C3, run by the engine each time**

One engine, oasis.py, runs the protocol for all three: founders on day 1, constitution on day 10, quiet days 11 to 20, one newcomer a day after that. Only the environment file changes: env_mult.py, env_ring.py, env_forms.py.

STAGE 1 · MULTIPLICATION

Every collapse lands on a prime

2	**3**	4	**5**	6	**7**	8	9	10	**11**	12	**13**
14	15	16	**17**	18	**19**	20	21	22	**23**	24	25
26	27	28	**29**	30	**31**	32	33	34	35	36	**37**
38	39	40	**41**	42	**43**	44	45	46	**47**	48	49
50	51	52	**53**	54	55	56	57	58	**59**	60	**61**
62	63	64	65	66	**67**	68	69	70	**71**	72	**73**
74	75	76	77	78	**79**	80	81	82	**83**	84	85
86	87	88	**89**	90	91	92	93	94	95	96	**97**
98	99	100	**101**	102	**103**	104	105	106	**107**	108	**109**
110	111	112	**113**	114	115	116	117	118	119	120	121

founders, day 1 · founding generators · collapse · built from generators

Numbers 12 to 121 arrive on days 21 to 130, one a day. A newcomer the generators cannot build pushes P above C = 0 and forces a collapse. The only fix that passes C1 to C3 is to make it a generator.

200,021 days

- 17,981 collapses. Every one logical: C1, C2 and C3 pass each time.
- The numbers that force them are exactly the primes from 13 to 200,009.
- No newcomer is ever built two ways. Unique factorization is the absence of illogical collapse.

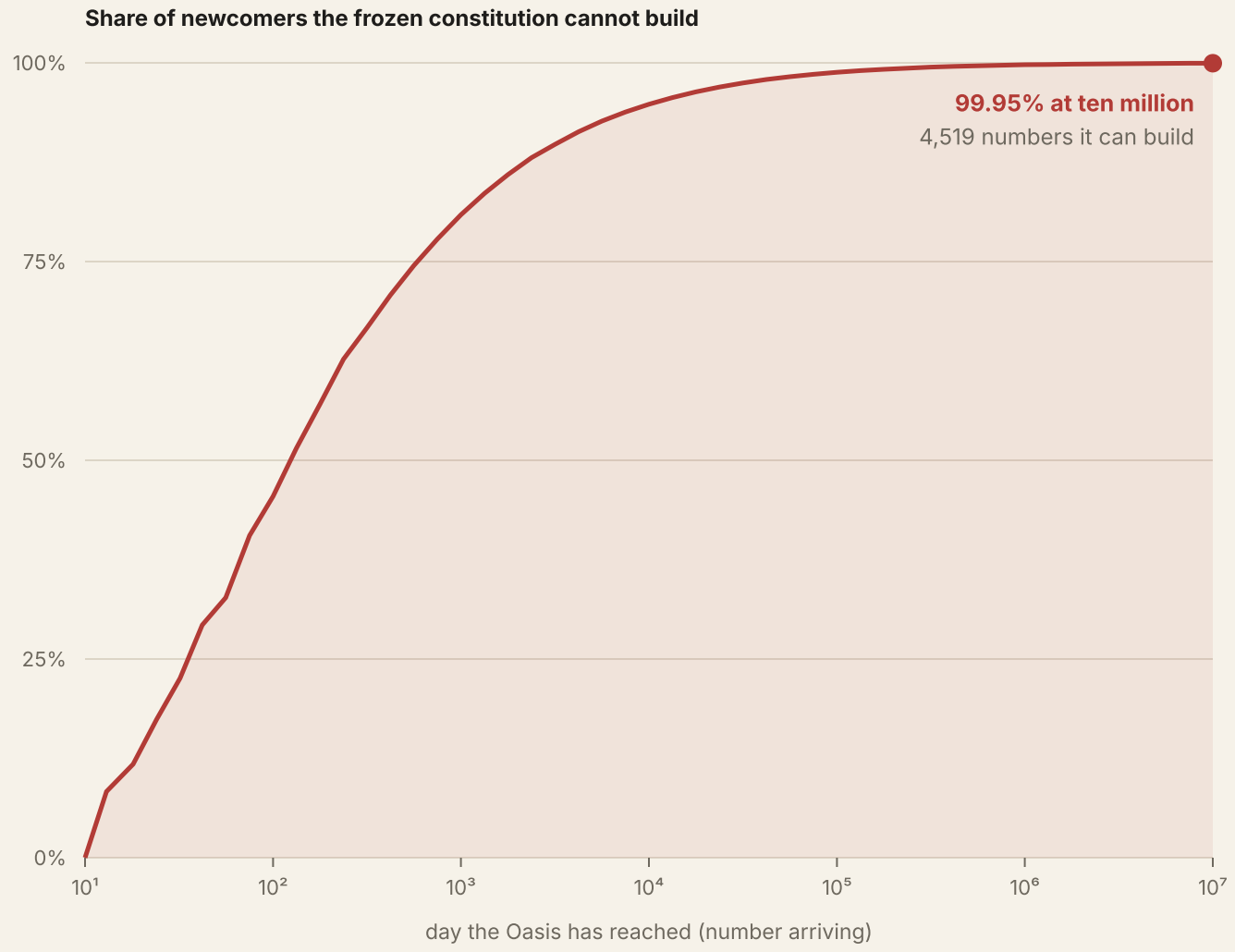
Raise the capacity, 20,000 days each

C	Collapses	Longest wait	Same primes added
0	2,258	0 days	✓
1	1,129	36 days	✓
3	564	66 days	✓
10	205	156 days	✓
30	73	378 days	✓
100	23	1,094 days	✓

A bigger C waits longer and collapses less often.
What it adds never changes: the primes.

EUCLID'S WALL

A constitution that stops collapsing dies



Freeze the constitution on day 10

- Generators stay 2, 3, 5, 7, 11. Nothing is ever added.
- The share it cannot build climbs toward all of them and never comes back.

The newcomer Euclid names

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 + 1 = 2311$$

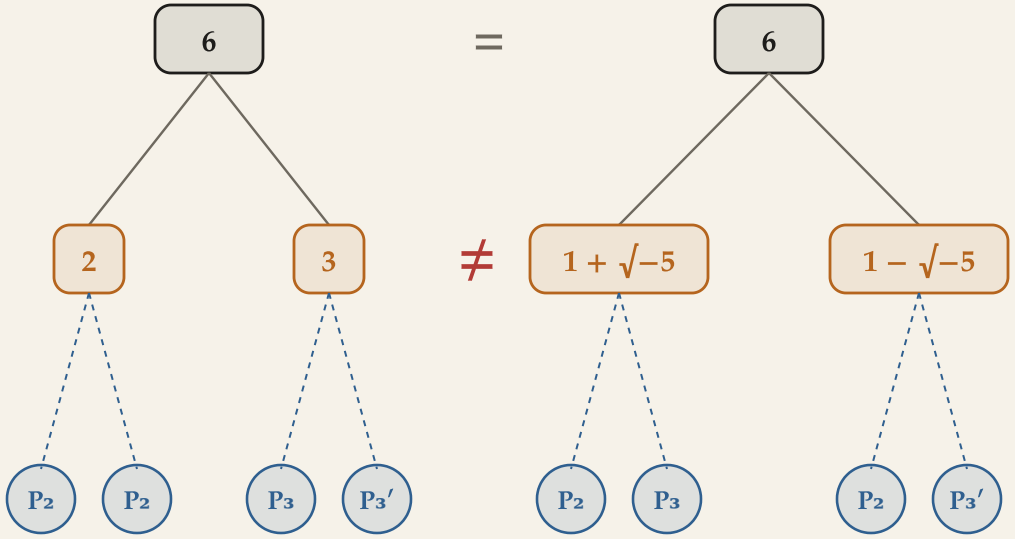
Every generator leaves remainder 1, so no generator divides it. Any finite list fails the same way. Elements IX.20, about 300 BC.

In collapse terms: a frozen constitution has no capacity large enough. It must collapse again and again, forever.

STAGE 2 · Z[√−5]

Day 33: the first illogical collapse

The newcomer 6 arrives built two ways, from atoms that cannot be split further.



Both lists use atoms, so no choice of atoms removes the conflict. The element language has no rule that passes C2. The engine ascends to ideals, where each atom splits and both sides agree:

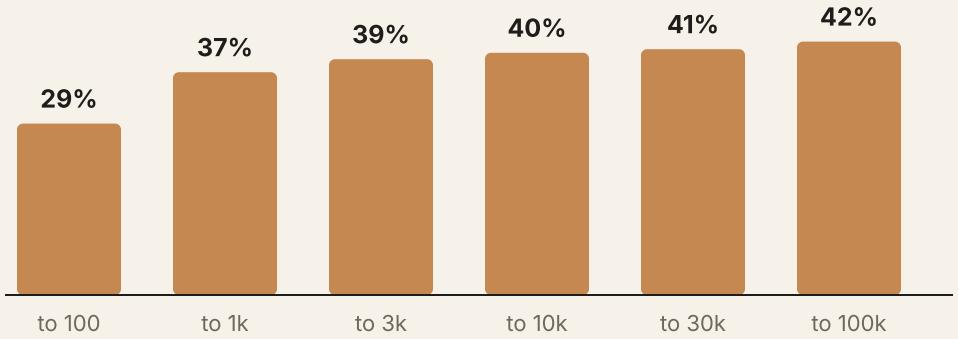
$6 = P_2 \cdot P_2 \cdot P_3 \cdot P_3'$ either way

$P_2 = (2, 1 + \sqrt{-5})$ $P_3 = (3, 1 + \sqrt{-5})$ $P_3' = (3, 1 - \sqrt{-5})$

Lamé claimed Fermat's Last Theorem in 1847 by assuming factoring is unique in rings like this. Liouville objected at once. Kummer had already seen it fail in 1844 and repaired it with ideal numbers in 1847. Dedekind turned those into ideals in 1871.

70,266 numbers a + b√−5, norm to 100,000

Share built more than one way, by size

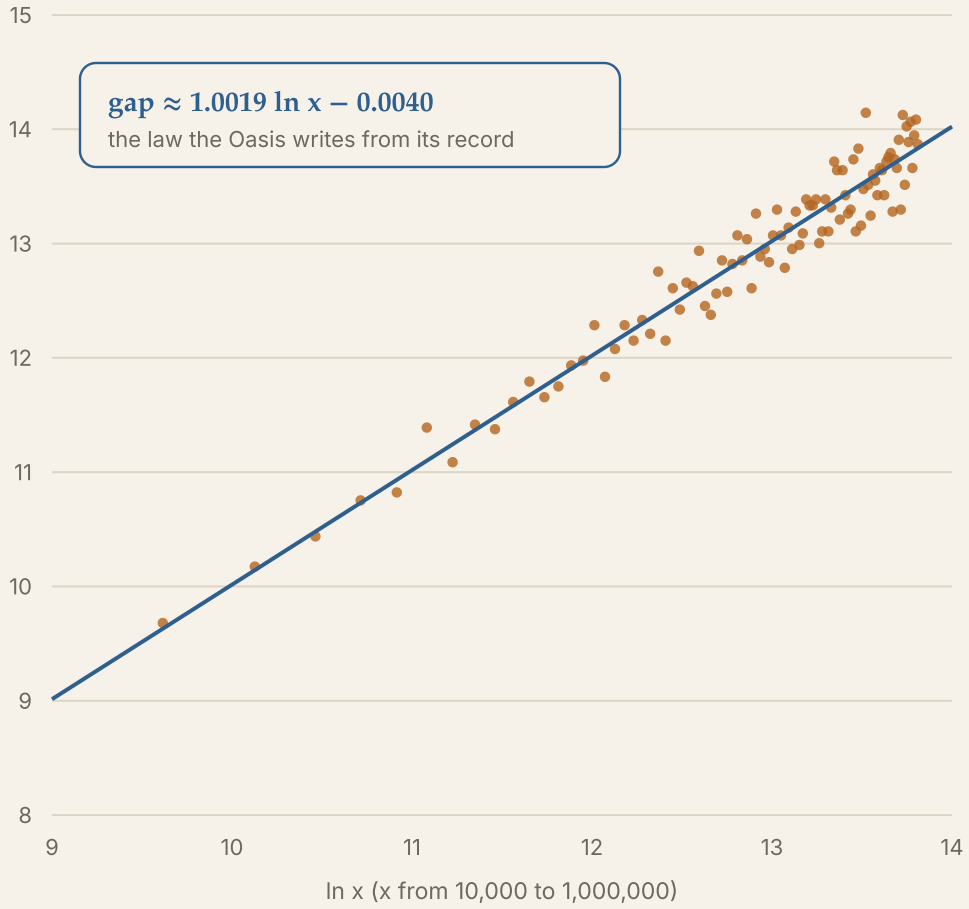


- After the ascent: no collapses in 70,243 more days.
- 117,146 factorizations checked against ideals, 0 disagreements.
- Lengths always agree, as Carlitz proved in 1960 for class number two.
- Most: 126 − 126√−5 has 36 different factorizations.

PREDICTION 1 · HOW MANY

The Oasis writes its own law, then forecasts blind

Average gap between collapses, blocks of 10,000 days



It then forecasts windows of a million numbers it has never seen

Window starts at	Forecast	Counted	Off by
10^{7}	61,753	**61,938**	0.30%
10^{8}	54,180	**54,208**	0.05%
10^{9}	48,171	**48,155**	0.03%
10^{10}	43,354	**43,427**	0.17%
10^{11}	39,412	**39,434**	0.06%
10^{12}	36,127	**36,249**	0.34%
10^{13}	33,348	**33,456**	0.32%
10^{14}	30,965	**30,892**	0.24%
10^{15}	28,901	**28,845**	0.19%
10^{16}	27,094	**27,168**	0.27%

error bar, ±0.5% full width

Fitted on days up to one million. Forecast out to ten quadrillion. Worst miss 0.34%.

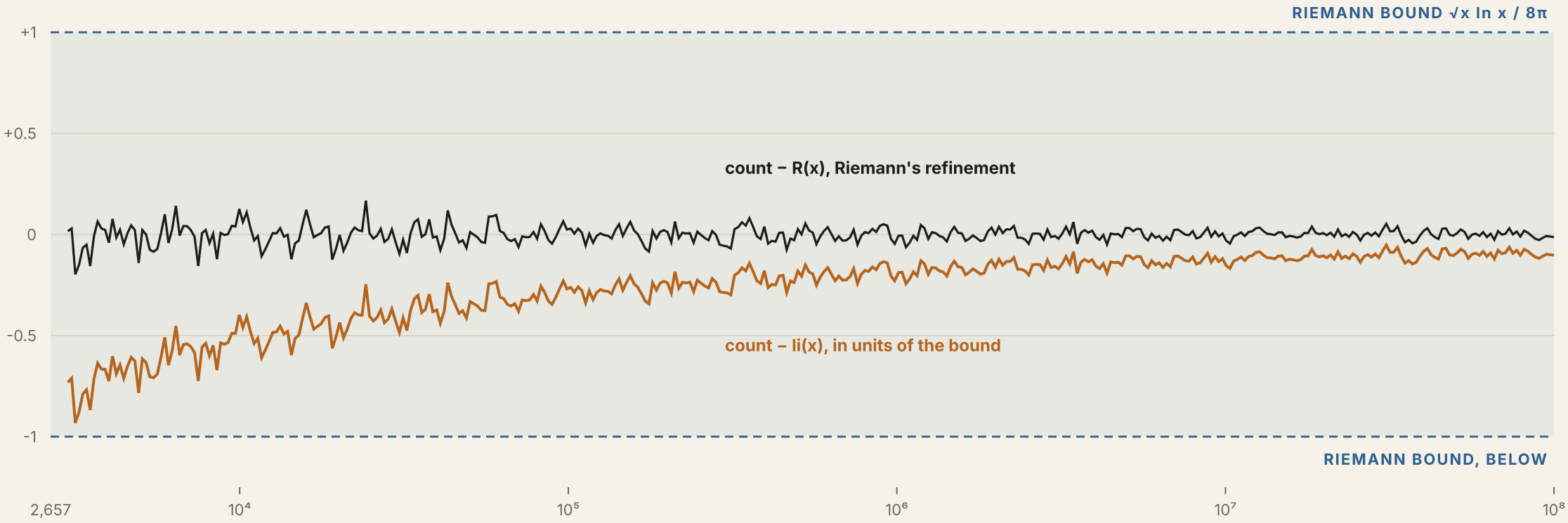
Gauss read the same law, gap about ln x, from prime tables as a teenager (letter to Encke, 1849).

PREDICTION 1 · HOW FAR OFF

How far the forecast can drift is the Riemann hypothesis

The law forecasts a count. The miss stays inside the band at all 5,761,072 primes checked, 2,657 to 10^8.

Closest approach: 0.98 of the bound, right below 2,659.

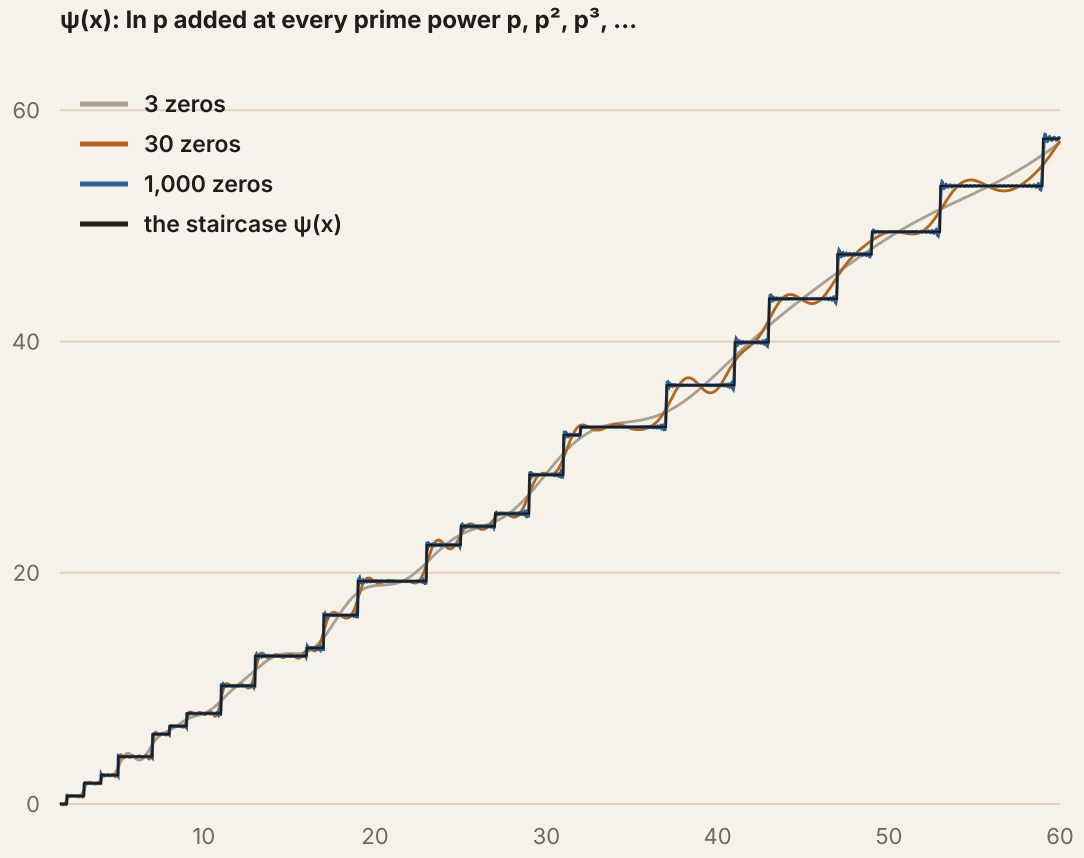


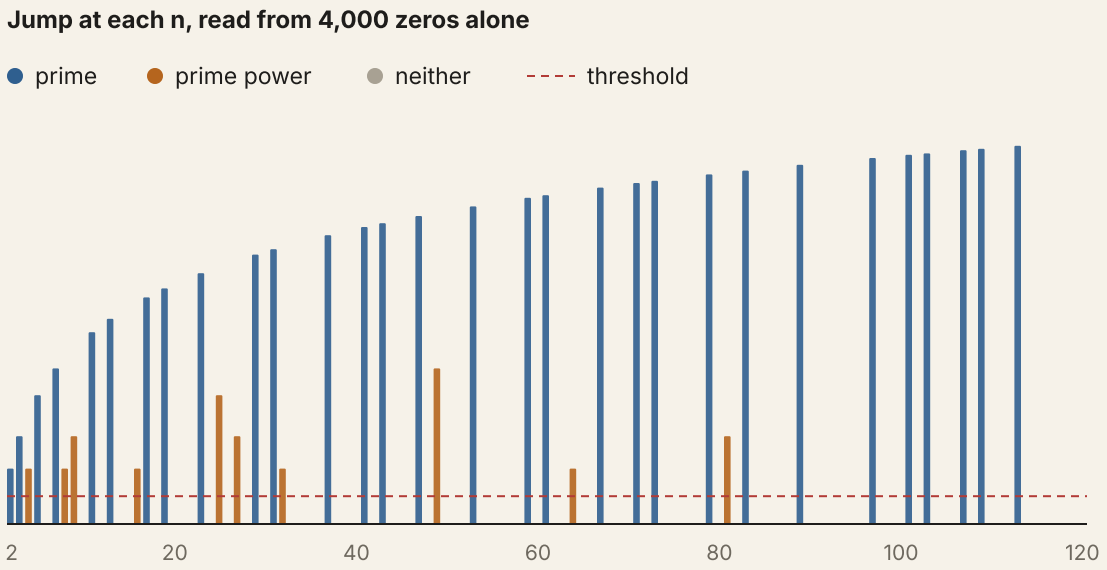
Schoenfeld (1976): the miss stays inside this band for every x ≥ 2,657 if and only if the Riemann hypothesis is true.

No number of days settles it from inside the Oasis. That is the first wall.

PREDICTION 2 · WHERE

The zeros of zeta know where every prime is





Zeros used	Every n read correctly up to	
100	39	12 primes, each told apart from prime powers
300	71	20 primes, each told apart from prime powers
1,000	179	41 primes, each told apart from prime powers
2,000	311	64 primes, each told apart from prime powers
3,000	431	83 primes, each told apart from prime powers
4,000	569	104 primes, each told apart from prime powers

No sieve, no division. Each zero is a wave; adding more of them sharpens the steps until every prime stands out.

Riemann's 1859 formula, in the form von Mangoldt proved in 1895. It predicts positions, and needs more zeros the further out you look.

Which primes are $x^2 + Dy^2$? Congruence laws, first

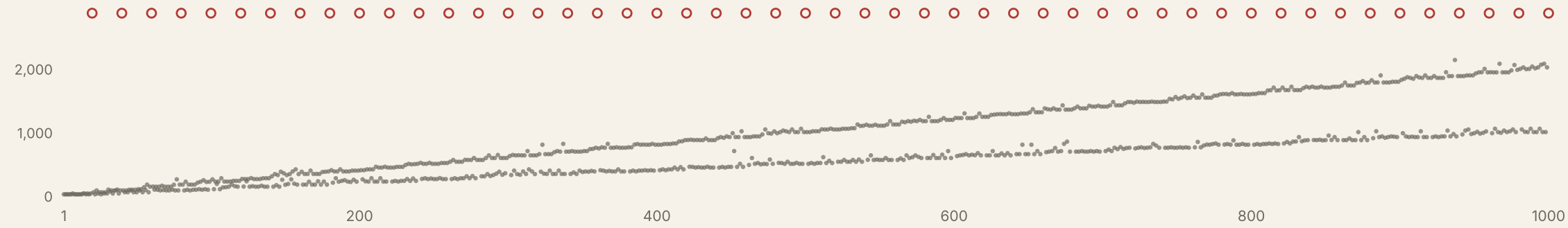
Each prime arrives carrying one fact: can it be written $x^2 + Dy^2$? The constitution must call it first.

D	Founders wrote	Collapses	Law it settles on	Primes since, all correct		History
1	mod 4	0	$p \equiv 1 \pmod 4$	78,487	✓	Fermat, 1640
2	mod 8	0	$p \equiv 1, 3 \pmod 8$	78,487	✓	Fermat
3	mod 3	0	$p \equiv 1 \pmod 3$	78,486	✓	Fermat
5	mod 7	1	$p \equiv 1, 9 \pmod{20}$	78,485	✓	Euler guessed, Lagrange proved

For D = 5 the founders see only one prime of the form, 29, and write a law mod 7. On day 21 the prime 41 breaks it. The engine collapses once, to mod 20, and that law then holds for every prime up to a million.

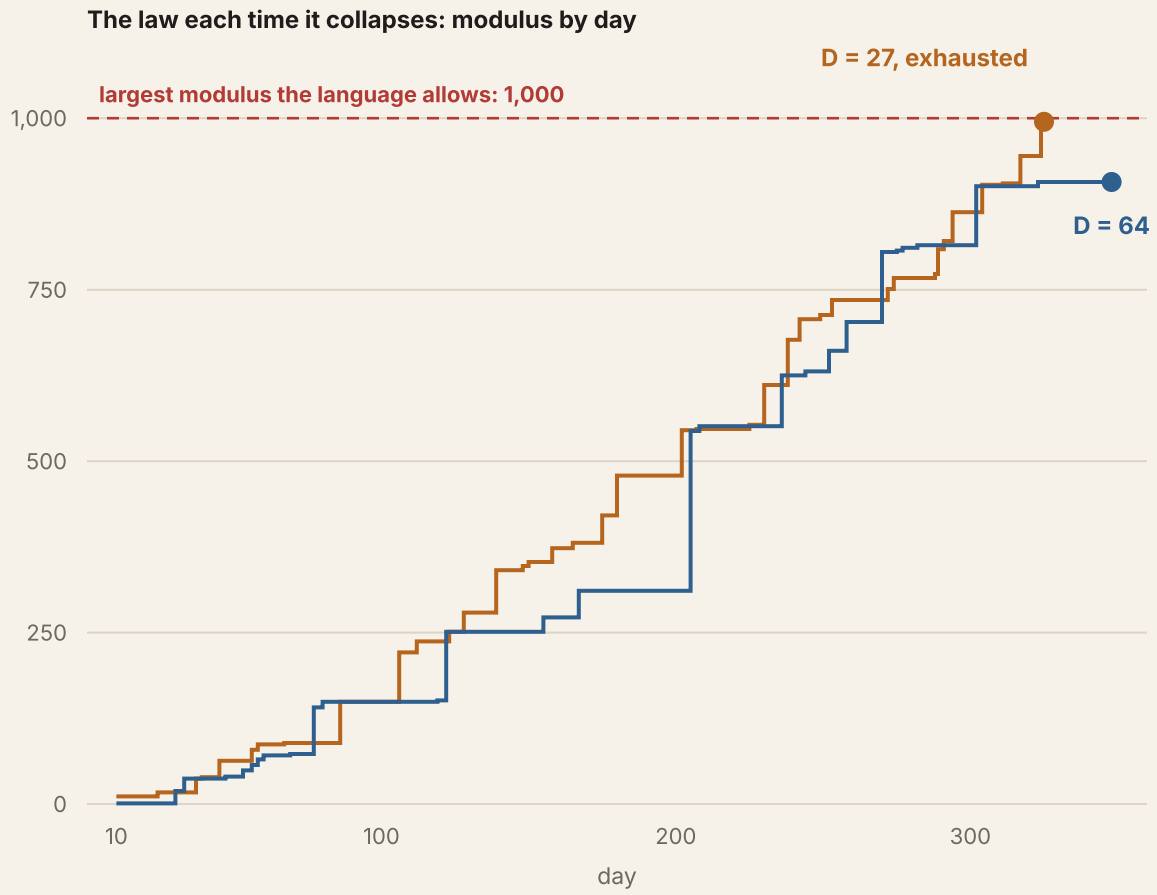
Every modulus up to 1,000 for D = 5, and the prime that refutes it. Lower dot: refuted sooner.

Red ring, top row: never refuted. Every multiple of 20.



STAGE 3 · THE LANGUAGE RUNS OUT

Day 325: no congruence can say it. A cubic can.



D = 27 · ascent to polynomials

$$p = x^2 + 27y^2 \Leftrightarrow x^3 - 2 \text{ has 3 roots mod } p$$

Found after 45 smaller polynomials failed on the record.
Then 78,181 more primes, to a million: 0 errors, 0 collapses.

Euler guessed it as "2 is a cube mod p". Gauss proved it with cubic reciprocity and never published the proof.

D = 64 · ascent to polynomials

$$p = x^2 + 64y^2 \Leftrightarrow x^4 - 2 \text{ has 4 roots mod } p$$

Found after 293 smaller polynomials failed on the record.
Then 78,159 more primes, to a million: 0 errors, 0 collapses.

Euler guessed it as "2 is a fourth power mod p". Gauss proved it in 1828, in his first memoir on biquadratic residues.

D = 27: 37 logical collapses, then on day 325 every modulus up to 1,000 is refuted.
D = 64: 26 logical collapses, exhausted on day 348. Class field theory says no modulus ever works.

STAGE 3 · LANGUAGE 3

A formula that prints the answer for every prime

A polynomial test checks one prime at a time. The third language prints every answer at once.

$$q \cdot \prod (1 - q^{6n})(1 - q^{18n}) = q - q^{7} - q^{13} - q^{19} + q^{25} + 2q^{31} - q^{37} + 2q^{43} - q^{61} \ldots$$

Multiply two of Euler's pentagonal series. No zeta, no fields, no decimals: only ±1 added at 256,795 places.

Rule: $p = x^2 + 27y^2$ exactly when the coefficient of q^p is 2.

Every prime p ≡ 1 (mod 3) below 200 and its coefficient

7	13	19	31	37	43	61	67	73	79	97	103	109	127	139	151	157	163	181	193	199
-1	-1	-1	2	-1	2	-1	-1	-1	-1	-1	-1	2	2	-1	-1	2	-1	-1	-1	-1

Blue: coefficient 2, and the prime is $x^2 + 27y^2$ ($31 = 2^2 + 27 \cdot 1^2$, $43 = 4^2 + 27 \cdot 1^2$, $109 = 1^2 + 27 \cdot 2^2$, ...). For p ≡ 2 (mod 3) the coefficient is 0.

Three rules, every prime from 5 to 1,000,000 (78,496 primes)

Search all x, y with $x^2 + 27y^2 = p$	**13,032**	brute force
2 is a cube mod p, $p \equiv 1 \pmod 3$	**13,032**	Euler, Gauss
Coefficient of q^p equals 2	**13,032**	pentagonal product

Same 13,032 primes, all three ways.

0 disagreements in 78,496. Coefficient −1: the other primes ≡ 1 (mod 3). Coefficient 0: p ≡ 2 (mod 3).

This is the shape of the Langlands program: a fact about primes that no congruence can see, read off a modular form.

The product η(6z)η(18z) is a weight-one modular form of level 108; this match is known, and the engine reached it on its own.

THE FINDING

One rule for every way to split 24

Split 24 into a + b. In $q \cdot \prod (1-q^{an})(1-q^{bn})$, the coefficient of q^p is 2 exactly when p is written by the simplest form of discriminant −ab.

p any prime not dividing 6ab. Simplest form: $x^2 + (ab/4)y^2$ when a is even, $x^2 + xy + ((ab+1)/4)y^2$ when a is odd.

a + b	ab	Prime p has coefficient 2 exactly when p =	First such primes	Primes with 2, to 10^6	Holds	Known as
1 + 23	23	$x^2 + xy + 6y^2$	59, 101, 167, 173, 211, 223	13,032	✓	van der Blij
2 + 22	44	$x^2 + 11y^2$	47, 53, 103, 163, 199, 257	13,058	✓	
3 + 21	63	$x^2 + xy + 16y^2$	67, 79, 127, 163, 277, 373	9,793	✓	
4 + 20	80	$x^2 + 20y^2$	29, 89, 101, 181, 229, 349	9,712	✓	
5 + 19	95	$x^2 + xy + 24y^2$	131, 239, 389, 419, 461, 821	4,848	✓	
6 + 18	108	$x^2 + 27y^2$	31, 43, 109, 127, 157, 223	13,032	✓	Gauss, cubic
7 + 17	119	$x^2 + xy + 30y^2$	263, 443, 557, 701, 1019, 1087	3,822	✓	
8 + 16	128	$x^2 + 32y^2$	41, 113, 137, 257, 313, 337	9,766	✓	
9 + 15	135	$x^2 + xy + 34y^2$	139, 151, 199, 331, 541, 619	6,504	✓	
10 + 14	140	$x^2 + 35y^2$	71, 149, 179, 331, 359, 379	6,511	✓	
11 + 13	143	$x^2 + xy + 36y^2$	179, 467, 653, 719, 797, 1013	3,897	✓	
12 + 12	144	$x^2 + 36y^2$	37, 61, 157, 193, 313, 349	9,760	✓	

941,939 checks, 0 failures. Why 24: the coefficient of q^n counts, with signs, the ways to write $24n = a \cdot u^2 + b \cdot v^2$ with u, v ≡ 5 (mod 6). Each row is a weight-one theta identity; I did not find the twelve stated as one rule. Verified, not proved.

THE LADDER

The engine climbs number theory and stops where we stop

Each rung is a collapse the engine hit by itself. Right: who hit the same wall first.

Day 22	**13 cannot be built** New generator. Primes never run out.	Euclid, about 300 BC
200,021 days	**nothing is built two ways** Every collapse logical.	Unique factorization, Gauss 1801
Day 33, Z[√−5]	**6 is built two ways** Contradiction. Ascent to ideals.	Kummer 1847, Dedekind 1871
Day 21, D = 5	**mod 7 breaks on 41** Logical collapse to mod 20.	Euler guessed, Lagrange proved
Day 325, D = 27	**every modulus refuted** Language exhausted. Ascent to $x^3 - 2$.	Euler guessed, Gauss proved
Language 3	**pentagonal products** One formula prints every answer.	Hecke, van der Blij 1952, Langlands

Wall 1 · how far the count can drift

Inside $\sqrt{x} \ln x / 8\pi$ forever is the Riemann hypothesis, open since 1859.

Wall 2 · a printing rule for every polynomial

Langlands reciprocity. Proven for $x^3 - 2$ and many more, open in general.

VERDICT

It predicts primes the way mathematics does, and no further

What it shows

- Primes are what multiplication must add to survive. Any capacity C adds the same ones.
- Unique factorization breaks in $Z[\sqrt{-5}]$ on day 33; ideals repair it with 0 contradictions after.
- A law fit on one million days forecasts collapse counts out to 10^{16} within 0.34%.
- 4,000 zeros of zeta rebuild every prime power up to 569 with no sieve.
- Congruence laws are a finite language. $x^2 + 27y^2$ and $x^2 + 64y^2$ exhaust it; polynomials take over.
- One pentagonal rule covers all twelve splits of 24, 941,939 checks, 0 failures.

What it cannot show

- A formula for the n-th prime that beats a sieve. Nothing here does that.
- That the forecast stays inside the band forever. That is the Riemann hypothesis.
- A printing rule for every polynomial. That is Langlands reciprocity, open in general.
- That the 24 rule is new. Each row is a theta identity a specialist may already know.
- A proof of the 24 rule. It is checked to a million, not proved.

Files: oasis.py is the engine. env_mult.py, env_ring.py and env_forms.py are the deserts. predict.py, zeros.py, eta.py and family_check.py run the forecasts and the checks.